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pqcrypto 1.0.0

Python bindings for post-quantum cryptography — ML-KEM, ML-DSA, SLH-DSA, Classic McEliece, HQC, and SNTRUP

```
pip install pqcrypto
```

Description Release files Security Release history

PQCrypto

build passing
python 3.9 | 3.10 | 3.11 | 3.12 | 3.13 | 3.14 pypi v1.0.0
license Apache-2.0

Python bindings for post-quantum cryptography. Wraps well-tested Rust implementations of NIST-standardized and candidate KEM and signature schemes via PyO3/maturin. All variants live in a single compiled wheel.

Installation

```
pip install pqcrypto
```

Or build from source:

Project links

Data verified by PyPI on Aug 15, 2026

Documentation
Homepage
Repository

Key dates

PyPI data

Released: Aug 15, 2026

LATEST RELEASE

1 maintainer

PyPI data

backbone

Credits

```
pip install maturin pytest
maturin develop
```

Quick Start

The Python API mirrors the Rust crate design: `keygen`, `encaps`, `decaps` (KEM) and `keygen`, `sign`, `verify` (signatures).

KEM — Key Encapsulation

```
from pqcrypto.kem.ml_kem_512 import keyge
from pqcrypto.kem.ml_kem_512 import Publi

pk, sk = keygen()
ct, ss = encaps(pk)
assert decaps(sk, ct) == ss

pk_obj = PublicKey(pk)
sk_obj = SecretKey(sk)
ct2, ss2 = pk_obj.encaps()
assert ss2 == sk_obj.decaps(ct2)
```

Signatures

```
from pqcrypto.sign.ml_dsa_44 import keyge
from pqcrypto.sign.ml_dsa_44 import Publi

pk, sk = keygen()
sig = sign(sk, b"message")
verify(pk, b"message", sig)  # returns No

# FIPS 204/205 context string:
sig = sign(sk, b"message", b"my-context")
verify(pk, b"message", sig, b"my-context"

# HashML-DSA / HashSLH-DSA pre-hash mode:
from pqcrypto import HashAlgorithm
sig = sign(sk, b"message", hash_algorithm
verify(pk, b"message", sig, hash_algorith

pk_obj = PublicKey(pk)
```

Author: Backbone Technologies Ltd.

GitHub Statistics

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- Repository
- Stars: **94**
- Forks: **28**
- Open issues: **1**
- Open PRs: **0**

Provenance

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Full provenance

All 8 files in this release have provenance metadata.

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License expression

Apache-2.0
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Requires

Python >=3.9

Provides Extra

`testing`

Tags

`post-quantum` `cryptography` `pqc` `ml-kem` `ml-dsa` `slh-dsa` `mceliece` `sntrup` `fips` `nist` `kem` `signature`

Classifiers

```
sk_obj = SecretKey(sk)
sig_obj = sk_obj.sign(b"message")
pk_obj.verify(b"message", sig_obj)
```

Size constants

```
from pqcrypto.kem.ml_kem_512 import PUBLI
from pqcrypto.sign.ml_dsa_44 import PUBLI
```

Algorithms

Modules live in `pqcrypto.kem` (KEM) and `pqcrypto.sign` (signatures).

Key Encapsulation

- **ML-KEM** (FIPS 203)
  - `ml_kem_512`, `ml_kem_768`, `ml_kem_1024`
- **Classic McEliece**
  - `mceliece_348864`
  - `mceliece_348864f`
  - `mceliece_460896`
  - `mceliece_460896f`
  - `mceliece_6688128`
  - `mceliece_6688128f`
  - `mceliece_6960119`
  - `mceliece_6960119f`
  - `mceliece_8192128`
  - `mceliece_8192128f`
- **SNTRUP**
  - `sntrup_653`
  - `sntrup_761`
  - `sntrup_857`
  - `sntrup_953`
  - `sntrup_1013`
  - `sntrup_1277`

Development Status
- 4 - Beta

Intended Audience
- Developers
- Science/Research

License
- OSI Approved :: Apache Software License

Natural Language
- English

Operating System
- MacOS
- Microsoft :: Windows
- POSIX :: Linux

Programming Language
- Python
- Python :: 3
- Python :: 3.9
- Python :: 3.10
- Python :: 3.11
- Python :: 3.12
- Python :: 3.13
- Python :: 3.14
- Rust

Topic
- Security :: Cryptography

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- **HQC** (FIPS 207)
  - `hqc_128`, `hqc_192`, `hqc_256`

Signatures

- **ML-DSA** (FIPS 204)
  - `ml_dsa_44`, `ml_dsa_65`, `ml_dsa_87`
- **SLH-DSA** (FIPS 205)
  - `slh_dsa_sha2_128s`
  - `slh_dsa_sha2_128f`
  - `slh_dsa_sha2_192s`
  - `slh_dsa_sha2_192f`
  - `slh_dsa_sha2_256s`
  - `slh_dsa_sha2_256f`
  - `slh_dsa_shake_128s`
  - `slh_dsa_shake_128f`
  - `slh_dsa_shake_192s`
  - `slh_dsa_shake_192f`
  - `slh_dsa_shake_256s`
  - `slh_dsa_shake_256f`

Security

The cryptographic implementations are provided by the backbone crates. These crates have not undergone a formal security audit; third-party review is recommended before production use. This Python wrapper is a thin shim — it validates key lengths, maps errors to Python exceptions, and provides idiomatic key objects.

License

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